

Mathematics

A Cultural Heritage

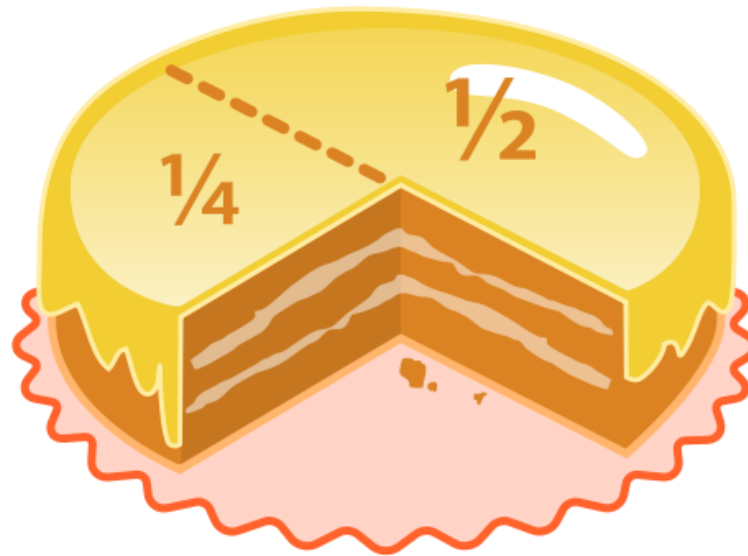
Lecture 2

Numbers – Part 4

Other Kinds of Numbers

Other Numbers: Fractions

- After counting, we learn to **share**



- **Fractions:** Whole number divided by another
- Form $\frac{m}{n}$, with m, n whole numbers

Fractions in Early Civilization

- Egyptians had **special fractions** (2000BC-1000BC)

$$1/5 = \overline{\text{||||}} \quad 1/360 = \overline{\text{𐎶𐎶𐎶}} \overline{\text{𐎵𐎵𐎵}}$$

- Greeks: Used *diacritic* mark (1000 years later)

- $\gamma' = \frac{1}{3} \quad \bar{\beta}\gamma' = \frac{2}{3}$

- $\mu\beta' = \frac{1}{42} \quad \text{or} \quad \mu\beta' = 40\frac{1}{2}$

- India: Similar to today (150BC – 500AD)

$$\overset{7}{-} \text{𑂔} = \frac{7}{15} = \frac{7}{15}$$

Decimal Fractions

- **Decimal fractions** (non-pos.): China (ca. 100BC)
- 920 – 980AD: in Middle East
- 12th Century: Iranian Mathematician used to approximate irrationals
- What are **irrational numbers**?

Ratios between Whole Numbers

- Pythagoras: strings of musical instruments with simple ratios of integers produce harmonious sound.



Commensurable Quantities

1. The Pythagoreans *believed* that all quantities are **commensurable**:

Given any two quantities (which can be represented as lengths of two line segments, or equivalently, two positive real numbers in today's terminology), there always exist a third quantity such that each of the two given quantities is a whole number multiple of the third quantity.

Commensurable Quantities

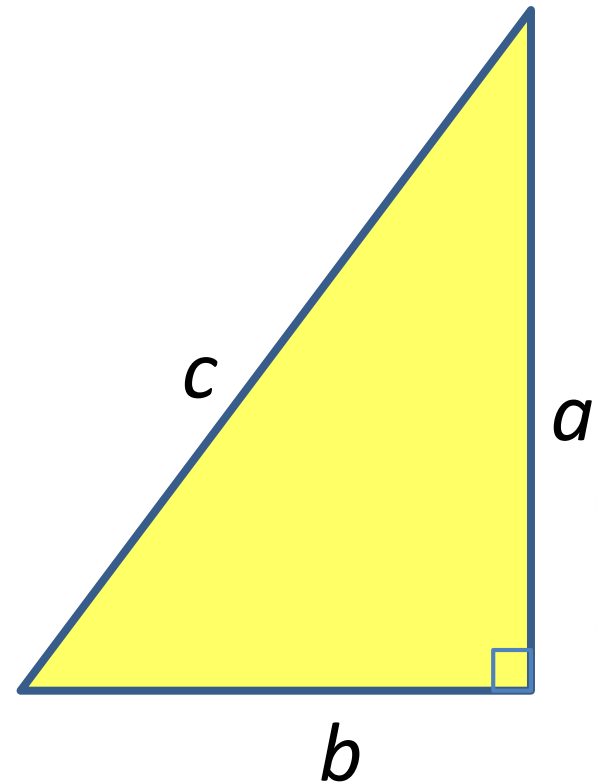


Pythagorean Triangles

- Pythagorean Theorem:

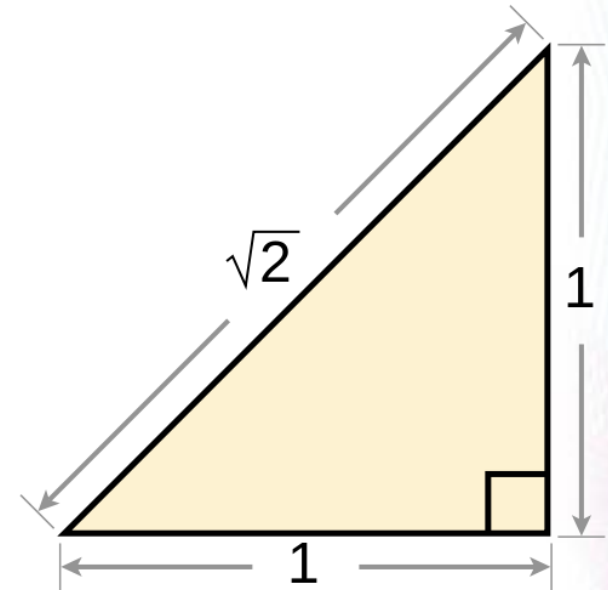
$$a^2 + b^2 = c^2$$

- Pythagoreans: b and c must be commensurable.



Downfall of a Dream

- Pythagoreans determine 1 and $\sqrt{2}$ **not** commensurable!
- Destroyed their philosophy of universe.
- Legend: **Hippasus** proved irrationality, **drowned at sea** (for discovery or for telling?).



Irrational Numbers

- More on discovery available in “Story of 1”
(17:15-21:00)



Irrationality of $\sqrt{2}$

- Suppose $\sqrt{2}$ rational, so $\sqrt{2} = \frac{m}{n}$, m, n integers.
Can assume m, n have no common factors
- Then $2 = \frac{m^2}{n^2}$, so $2n^2 = m^2$.
- Thus m is even (call $m = 2k$). But then $n^2 = 2k^2$.
- So n is even. This is absurd; m, n have no common factors! So no m, n could possibly exist!

Proof by Contradiction

- This strategy of proof is called *Reductio ad absurdum* (proof by contradiction).
- It is a very powerful technique based on logic and deductive reasoning.

Imaginary Numbers

- Square root of -1.
- René Descartes coined
- Usually write

$$i = \sqrt{-1}$$

- Not real number (squares always non-negative).

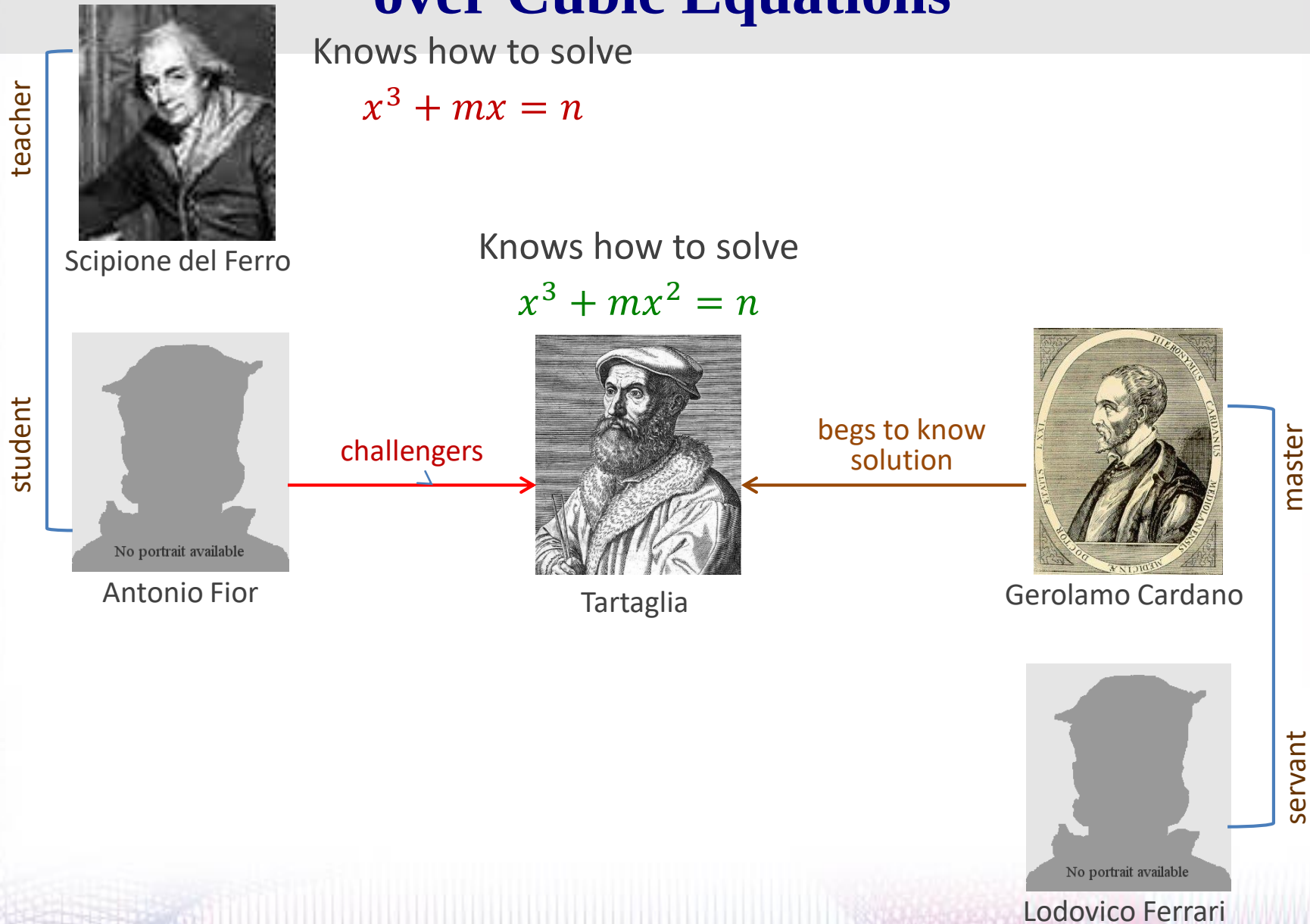
Complex Numbers

- Add multiples of i to real numbers:
 - $2 - 3i$
 - $4.3 + 0.21i$,
 - $\sqrt{2} i (= 0 + \sqrt{2}i)$
 - $a + bi$, a, b real

Quadratic Equations

- First exposure to complex numbers:
 - Solutions to $ax^2 + bx + c = 0$:
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
 - When $b^2 - 4ac < 0$, get complex number
- Historically: from **cubic equations**
- Many ancient civilizations solved quadratic equations

16th Century Italy: Challenge/Feud over Cubic Equations



Solution of Cubic Equations

- Cardano, *Ars Magna* (*The Great Art*) 1545:
 - General cubic equation reduced to $x^3 = 3px + 2q$
 - Solution is given by $x = (q + w)^{\frac{1}{3}} + (q - w)^{\frac{1}{3}}$, for $w = (q^2 - p^3)^{\frac{1}{2}}$.
- $q^2 \geq p^3$: just one real solution x .
- $q^2 < p^3$: three real roots, but w is square root of negative number!
- **Rafael Bombelli** (1526-1572), *L'Algebra* (1572):
 - Allow complex numbers, gives all three real solutions.
 - Imaginary parts of $(q + w)^{\frac{1}{3}}$ and $(q - w)^{\frac{1}{3}}$ cancel

Complex Numbers as a Tool

- First time historically complex numbers accepted:
 - Only as tool for real solutions
- Formula of Cardano is:

$$x = \sqrt[3]{q + \sqrt{q^2 - p^3}} - \sqrt[3]{q - \sqrt{q^2 - p^3}}$$

This sort of formula is called solution in radicals.

- **Lodovico Ferrari** (student of Cardano):
 - Solution in radicals for the **quartic** (i.e., degree-4) **equations** in 1540 (published also in *Ars Magna*)

Doubt about Complex Numbers

- Useful for cubic/quartic equations
 - But no physical interpretation!
 - Imaginary numbers and complex numbers looked doubtful
- Even two centuries after Bombelli's *L'Algebra*, **Euler** wrote in his *Algebra* (1770) :

“All such expressions as $\sqrt{-1}$, $\sqrt{-2}$, etc., are consequently impossible or imaginary numbers, since they represent roots of negative quantities; and of such numbers we may truly assert that they are neither nothing, nor greater than nothing, nor less than nothing, which necessarily constitutes them imaginary or impossible.”

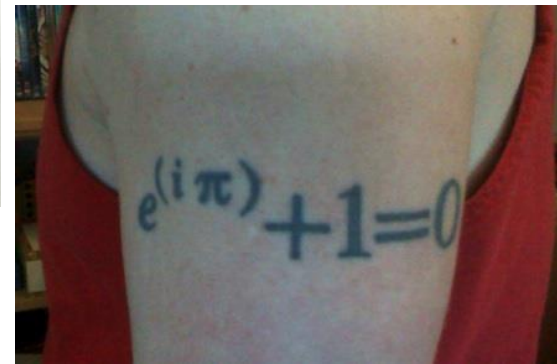


Leonhard Euler (1707-1783)
[source: wikipedia]

Complex Numbers

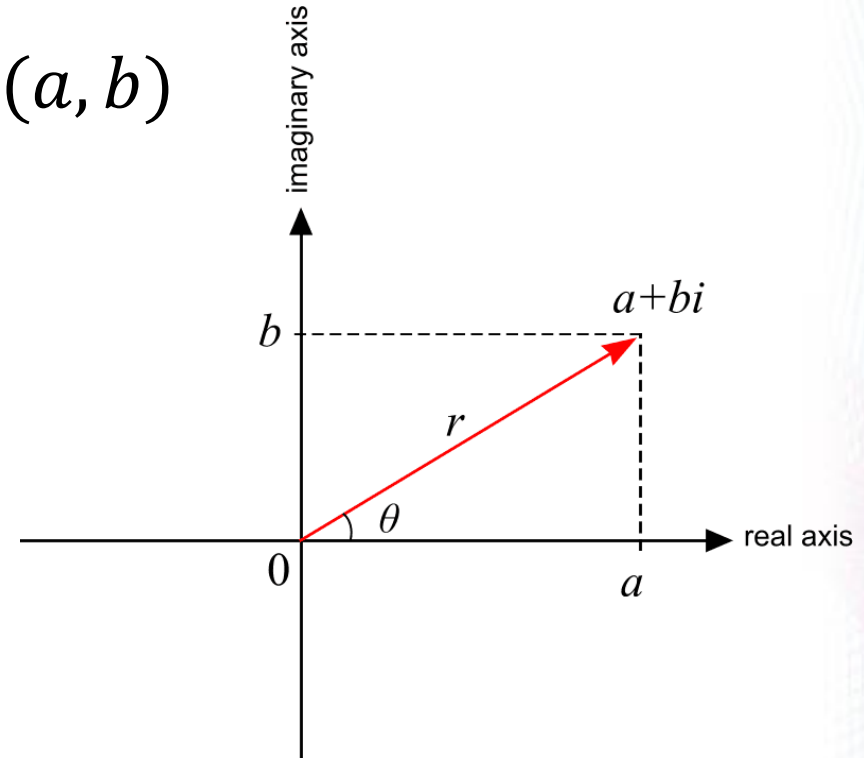
- But Euler used complex numbers!

$$e^{\pi\sqrt{-1}} + 1 = 0$$



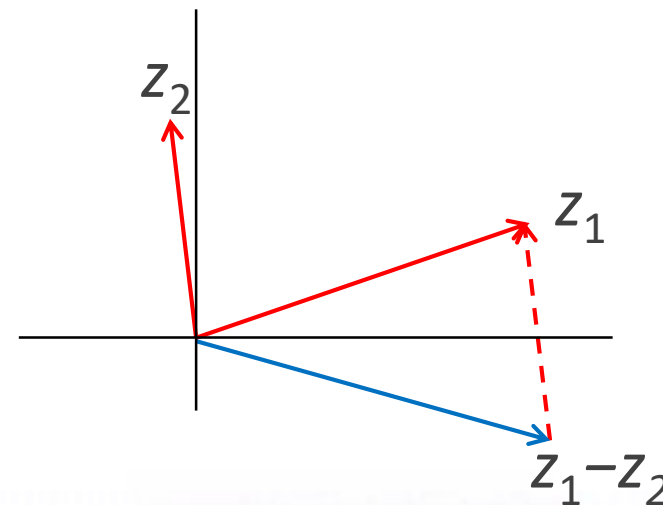
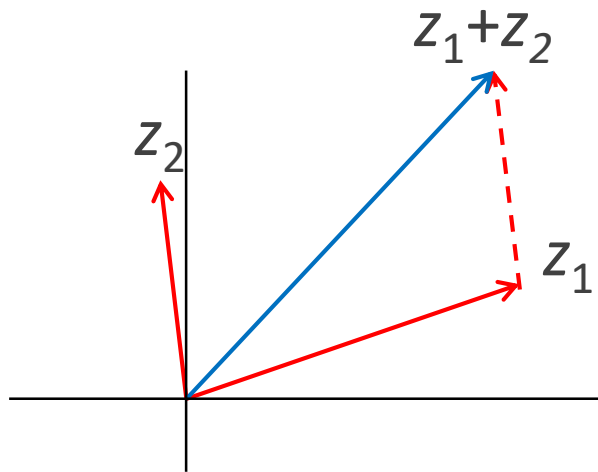
Geometric Interpretation of Complex Numbers

- **Caspar Wessel** (1745-1818), Norwegian–Danish:
 - Geometric interpretation
 - $a + bi$: line from $(0,0)$ to (a, b)
- Also **polar coordinates**:
 - Length r
 - Angle θ



Geometric Interpretation of Complex Numbers

- Wessel also discovered:
 - **Addition** and **subtraction**: Join the corresponding arrows
 - **Multiplication**: Multiply r_1, r_2 , add θ_1, θ_2 .



Applications of Complex Numbers

- **Fundamental Theorem of Algebra:**

Any degree- n polynomial equation

$$x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0$$

has exactly n complex solutions (counting multiplicities)

- **Laplace Transform:**

- Takes real-valued function $f(t)$ to complex-valued function $F(s)$:

$$F(s) = \mathcal{L}\{f(t)\}(s) := \int_0^{\infty} e^{-st} f(t) dt$$

- Used by engineers quite a lot

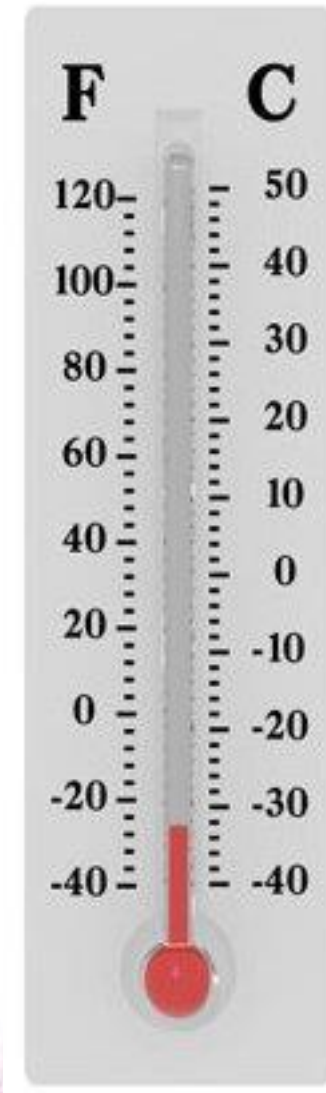
Negative Numbers

- **Negative numbers** first appear in a Chinese mathematical treatise *The Nine Chapters on the Mathematical Art* (Jiu Zhang Xuan Shu 九章算術, 2nd Century BC – 1st Century AD)
- First indicated **debts**.
- Many claimed negative numbers were absurdity

Negative Numbers

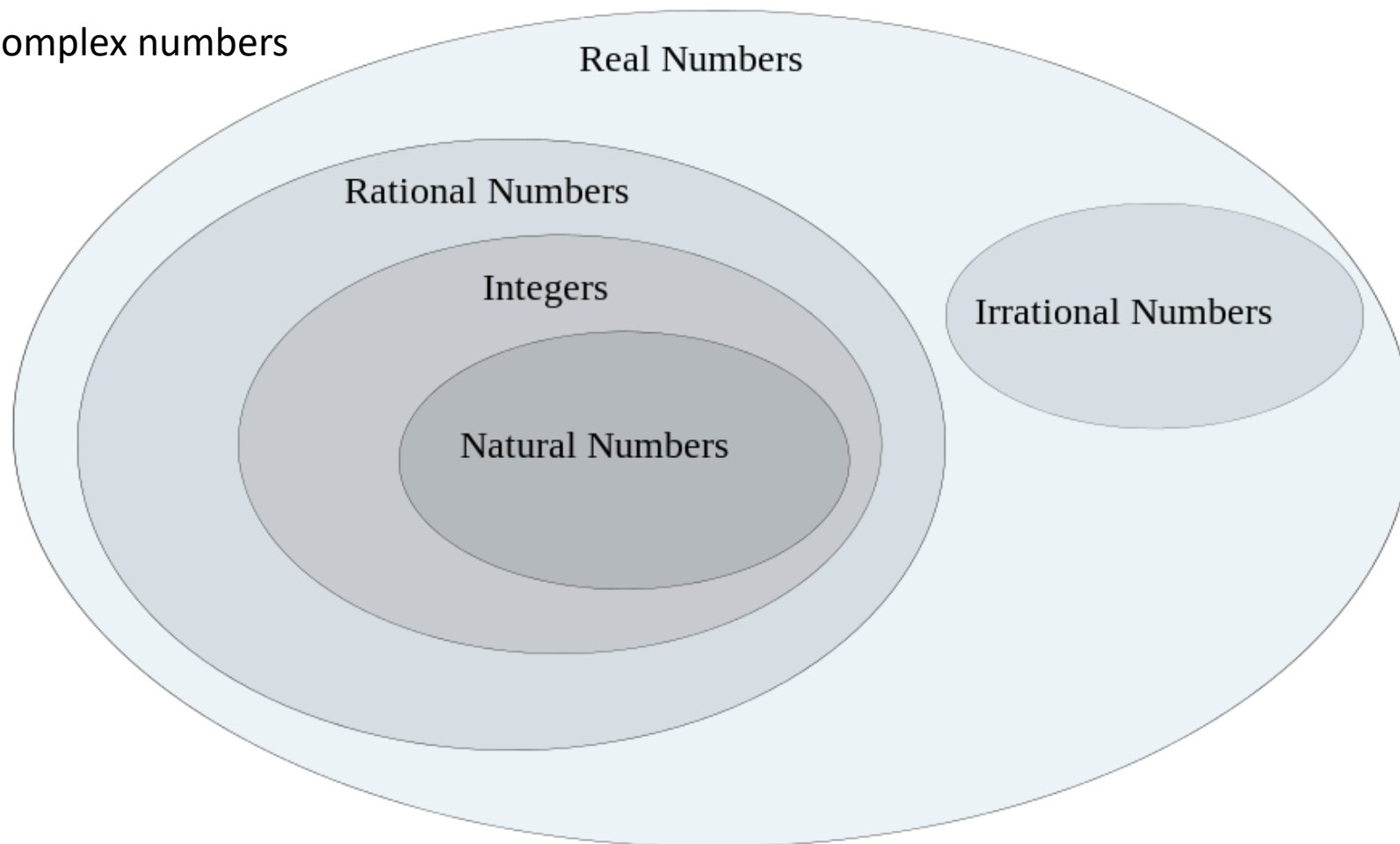
- However, occur often in daily life:
 - Temperatures below freezing
 - Bank info

Bank Account Information		
Starting Balance	01/12/2014	23000
Deposit	07/12/2014	1100
Withdrawal	13/12/2014	1300
New Balance	01/01/2015	22800



Families of Numbers

Complex numbers



"Real numbers" by Damien Karras - Own work.
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Hypercomplex Numbers

- Higher dimensional analogues?
- 3-dimensional version:
 - Arithmetic is 3-D rotations
 - $a + bi + cj, i^2 = j^2 = -1.$
 - Unsuccessful!
- Breakthrough: 16th October, 1843.

Quaternions

1. **William Hamilton** (1805–1865, Ireland) working hard looking for 3-D analogue.
2. 16 October 1843: Walk with his wife along the Royal Canal in Dublin; answer suddenly occurred to him: Try **4-dimensions!**
3. Carved equations with a penknife into the side of the nearby Broom Bridge:

$$i^2 = j^2 = k^2 = ijk = -1$$



William Hamilton (1805–1865)
[source: wikipedia]



Quaternions

- Numbers of form $a+bi+cj+dk$. Called **quaternions**.
- Basic imaginary units i, j, k satisfy:
$$ij = k = -ji, \quad jk = i = -kj, \quad ki = j = -ik,$$
- Multiplication doesn't **commute**!
- Applications to quantum mechanics, 3-D animation, electro-mechanics, ...